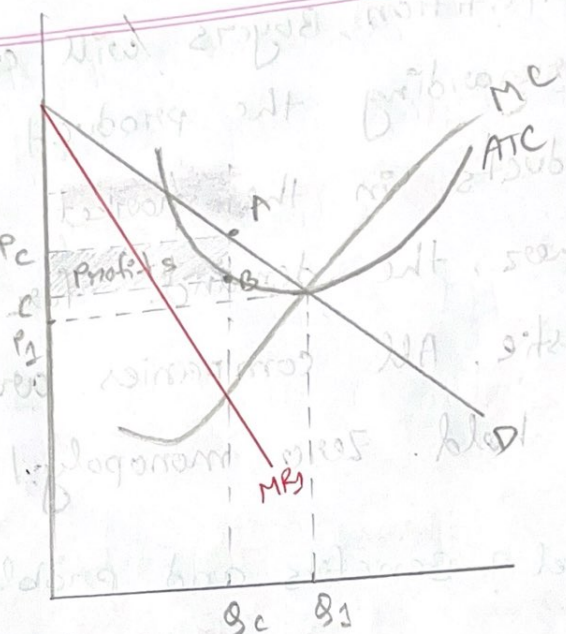


## 10. What is cartel? Benefits and problems?

Cartel is an organization of firms that reduces output and increases price in an effort to increase joint profits. In this theory of oligopoly, oligopolistic firms act as if there were only one firm in the industry. In short, they form a cartel to capture the benefits that would exist for a monopolistic.

### Benefits of cartel:

Forming and maintaining cartel has its benefits.



This figure shows an industry in the long-run competitive equilibrium. Price is  $P_1$  and quantity of output is  $Q_1$ . The industry is producing the output at which  $P = MC$ , that means, there is zero economic profit.

Now, suppose that the firms making up the industry form a cartel and reduces output to  $Q_c$ . The new price is  $P_c$  and there are profits equal to the area  $CP_cAB$  which can be shared among the members of the cartel. With no cartel there is

no profits, with a cartel, profits are earned. Thus the firms have an incentive to form a cartel and to behave cooperatively rather than competitively.

However firms may not be able to form a cartel, even though they have a profit incentive to do so.

Also even if they are able to form the cartel, the firms may not be able to maintain it. Firms that wish to form and maintain a cartel will encounter several problems, like

### 1. The problem of forming a cartel:

#### 1. Legal prohibition:

Legislation prohibits certain types of cartels in many countries like US. Organizing and forming a cartel is also costly.

#### 2. The problem of forming the cartel:

Even if it were legal, getting the sellers of an

Industry together to form a cartel can be costly, even when the number of sellers is small. Each potential member has an incentive to be free rider to stand by and take a free ride on the actions of others.

### 3. The problem of formulating cartel policy:

This problem occurs when there are as many policy proposals as there are cartel members. Reaching agreement may be difficult. Such disagreements become harder to resolve as the differences among cartel members in costs, size and so forth grow.

### 4. The problem of entry into the industry:

Even if the cartel members manage to agree on a policy that generates high profits, those high profits will provide an incentive for firms outside the industry to join the industry. If

current cartel from entering

### 5. The problem

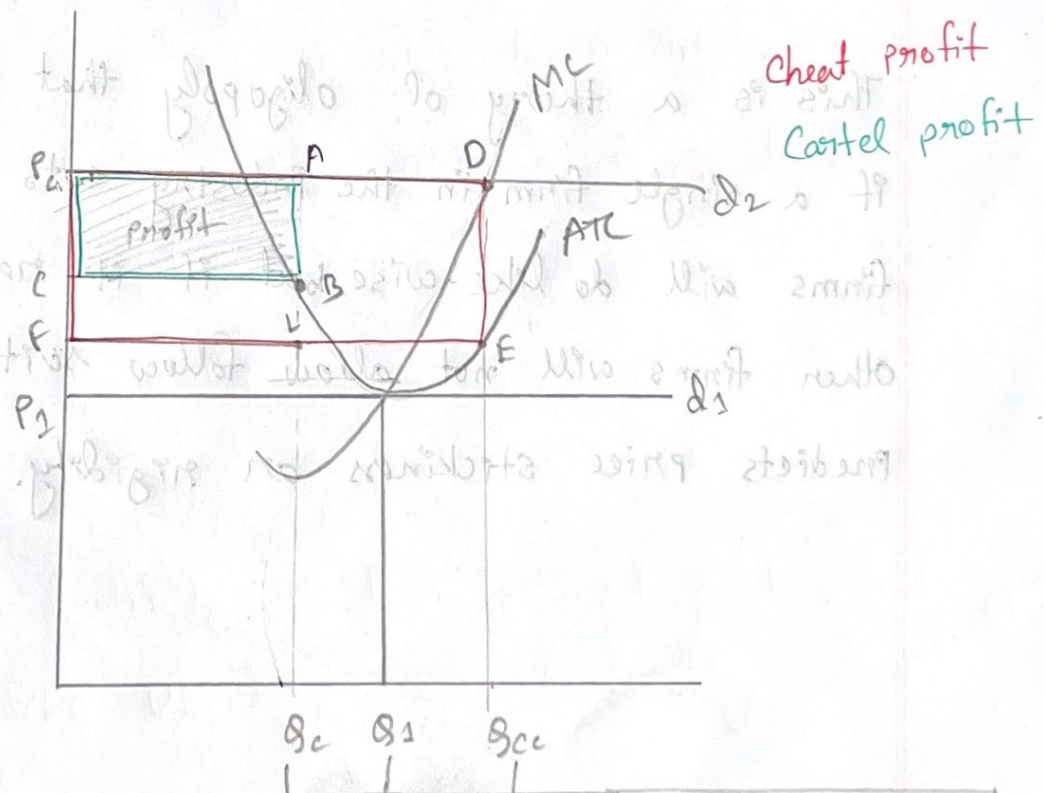
As paradoxical agreement is on it.

Quantity firm produces when it adheres to cartel agreement  
profits =  $eP_cAB$

current cartel members cannot keep new suppliers from entering, the cartel is likely to break up.

### 5. The problem of cheating:

As paradoxical as it first appears, after the cartel agreement is made, members have an incentive to cheat on it.



Quantity firm ✓  
Produces when it adheres to cartel agreement  
profits =  $CP_{cAB}$

Quantity firm  
Produces when there is no cartel agreement  
profits = 0

Quantity firm ✓  
Produces when it cheats on the cartel agreement, but other firms do not  
Profit =  $FP_{cDE}$

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If this firm can cheat on the cartel agreement, so can others. Given the monetary benefits gained by cheating, it is likely that the cartel will exist for only a short time.

13. Let us consider a duopoly market where two firms produce a homogenous product. The demand curve faced by the firm is  $P = 100 - 0.5(q_1 + q_2)$ . The cost functions are  $C_1 = 5q_1$ ,  $C_2 = 0.5q_2$

i) Find the reaction functions for both firms and find Cournot solution.

ii) Find the quasi competitive and collusion solutions and compare between them.

i) Given that

$$P = 100 - 0.5(q_1 + q_2)$$

$$C_1 = 5q_1$$

$$C_2 = 0.5q_2$$

$$\text{So, } MC_1 = 5$$

$$\text{So, } MC_2 = q_2$$

$$\text{Now, } TR_1 = Pq_1$$

$$= \{100 - 0.5(q_1 + q_2)\} q_1$$

$$= 100q_1 - 0.5q_1^2 - 0.5q_1q_2$$

$$\therefore MR_1 = 100 - q_1 - 0.5q_2$$

At profit maximizing level -

$$MR_1 = MC_1$$

$$\therefore 100 - q_1 - 0.5q_2 = 5$$

$$\Rightarrow q_1 = 100 - 5 - 0.5q_2$$

$$\Rightarrow q_1 = 95 - 0.5q_2 \quad \text{--- (i)}$$

This is the firm-1's reaction curve.

$$\text{Again, } TR_2 = \cancel{MR_2} P q_2$$

$$= \{100 - 0.5(q_1 + q_2)\} q_2$$

$$= 100q_2 - 0.5q_1q_2 - 0.5q_2^2$$

$$\therefore MR_2 = 100 - 0.5q_1 - q_2 \quad \text{---}$$

At profit maximizing level

$$MR_2 = MC_2$$

$$\therefore 100 - 0.5q_1 - q_2 = q_2$$

$$\Rightarrow 100 - 0.5q_1 = 2q_2$$

$$\Rightarrow q_2 = 50 - 0.25q_1 \quad \text{--- (ii)}$$

this is the firm-2's reaction function.

Now, substituting the value of  $q_2$  from (ii) into

$$q_1 = 95 - 0.5(50 - 0.25q_1)$$

$$\Rightarrow q_1 = 95 - 25 + 0.125q_1$$

$$\Rightarrow q_1 - 0.125q_1 = 70$$

$$\Rightarrow 0.875q_1 = 70$$

$$\Rightarrow q_1 = \frac{70}{0.875}$$

$$= 80$$

$$\therefore q_2 = 50 - 0.25 \times 80$$

$$= 30$$

$$\text{So, } P = 100 - 0.5(q_1 + q_2)$$

$$= 100 - 0.5(80 + 30)$$

$$= 45$$

$$\pi_1 = Pq_1 - C_1$$

$$= Pq_1 - 5q_1$$

$$= (45 \times 80) - 5(80)$$

$$= 3600 - 400$$

$$= 3200$$

$$\pi_2 = Pq_2 - 0.5q_2$$

$$= (45 \times 30) - 0.5(30)$$

$$= 900$$

This is Cournot solution.

ii) For quasi competitive solution:

$$P = MC$$

therefore,  $P = MC_1$  for firm-1

$$\Rightarrow 100 - 0.5(q_1 + q_2) = 5$$

$$\Rightarrow 100 - 0.5q_1 - 0.5q_2 = 5$$

$$\Rightarrow 95 = 0.5(q_1 + q_2)$$

$$\Rightarrow q_1 + q_2 = 190 \quad \text{--- (iii)}$$

therefore,  $P = MC_2$  for firm-2

$$100 - 0.5(q_1 + q_2) = q_2$$

$$\Rightarrow 100 - 0.5q_1 - 0.5q_2 = q_2$$

$$\Rightarrow 100 - 0.5q_1 = q_2 + 0.5q_2$$

$$\Rightarrow 100 - 0.5q_1 = 1.5q_2$$

$$\Rightarrow q_2 = \frac{100 - 0.5q_1}{1.5} \quad \text{--- (iv)}$$

Substituting the value of  $q_2$  from (iv) into (iii)

$$q_1 + \frac{100 - 0.5q_1}{1.5} = 190$$

$$\Rightarrow \frac{1.5q_1 + 100 - 0.5q_1}{1.5} = 190$$

$$\Rightarrow q_1 + 100 = 285$$

$$\Rightarrow q_1 = 185$$

$$\text{So, } q_2 = \frac{100 - 0.5(185)}{1.5} = 5$$

$$\text{Thus, } P = 100 - 0.5(185 + 5) = 5$$

$$\pi_1 = Pq_1 - 5q_1$$

$$= 5(185) - 5(185)$$

$$= 0$$

$$\pi_2 = Pq_2 - 0.5q_2^2$$

$$= (5 \times 5) - 0.5(5)^2$$

$$= 12.5$$

Thus competitive solution is

$$q_1 = 185, q_2 = 5, P = 5, \pi_1 = 0, \pi_2 = 12.5$$

Collusion solution:

If firms collude then,

$$\pi = \pi_1 + \pi_2$$

$$= Pq_1 - c_1 + Pq_2 - c_2$$

$$= (100 - 0.5q_1 - 0.5q_2)q_1 - 5q_1 +$$

$$(100 - 0.5q_1 - 0.5q_2)q_2 - 0.5q_2^2$$

$$\begin{aligned}
 &= 100q_1 - 0.5q_1^2 - 0.5q_1q_2 - 5q_1 + 100q_2 - 0.5q_1q_2 - 0.5q_2^2 - 0.5q_2^2 \\
 &= 100q_1 - 0.5q_1^2 - 0.5q_1q_2 - 0.5q_1q_2 - 0.5q_2^2 + 100q_2 - 0.5q_2^2 - 5q_1 \\
 &= 100(q_1 + q_2) - 0.5(q_1^2 + 2q_1q_2 + q_2^2) - 5q_1 \\
 &= 100(q_1 + q_2) - 0.5(q_1 + q_2)^2 - 5q_1
 \end{aligned}$$

Setting the partial derivatives of  $\pi$  equal to zero

$$\frac{\partial \pi}{\partial q_1} = 100 - q_1 - 5 - q_2 = 0$$

$$\Rightarrow 95 - q_1 - q_2 = 0$$

$$\therefore q_1 = 95 - q_2$$

$$\frac{\partial \pi}{\partial q_2} = 100 - q_2 - q_1 = 0$$

$$\frac{\partial \pi}{\partial q_1} = 100 -$$

$$\Rightarrow 100$$

$$\Rightarrow$$

$$\frac{\partial \pi}{\partial q_2} = 100 -$$

$$\Rightarrow 100$$

$$\Rightarrow$$

$$\Rightarrow$$

$$\Rightarrow$$

$$\therefore q_1 = 95 - q_2$$

$$\therefore P = 100 - 0.5C$$

$$\pi_1 = Pq_1 - 5q_1$$

$$\pi_2 = Pq_2 - 0.5q_2$$

This is the

$$\frac{\partial \pi}{\partial q_1} = 100 - q_1 - 0.5q_2 - 0.5q_2 - 5 = 0$$

$$\Rightarrow 100 - 5 - q_1 - q_2 = 0$$

$$\Rightarrow 95 - q_2 = q_1$$

$$\therefore q_1 = 95 - q_2$$

$$\frac{\partial \pi}{\partial q_2} = 100 - 0.5q_1 - 0.5q_1 - q_2 - q_2 = 0$$

$$\Rightarrow 100 - 2q_2 - 0.5q_1 - 0.5q_1 = 0$$

$$\Rightarrow 100 - 2q_2 - q_1 = 0$$

$$\Rightarrow 100 - 2q_2 - 95 + q_2 = 0 \quad [\because q_1 = 95 - q_2]$$

$$\Rightarrow 5 - q_2 = 0$$

$$\therefore q_2 = 5$$

$$\therefore q_1 = 95 - q_2 = 95 - 5 = 90$$

$$\therefore P = 100 - 0.5(90 + 5) = 52.5$$

$$\pi_1 = Pq_1 - 5q_1 = 52.5(90) - 5(90) = 4275$$

$$\pi_2 = Pq_2 - 0.5q_2^2 = (52.5 \times 5) - 0.5(5)^2 = 250$$

This is the collusion solution.

Here, competitive solution is better for consumers because the price is lower than the collusion solution. So, they will maximize the quantity. But, for producers collusion solution is better because the price is higher than the competitive solution. So, they will maximize their profit.

14. Suppose, there are two firms in a duopoly market. Firm-1's demand function is  $Q_1 = 12 - 2P_1 + P_2$  and firm-2's demand function  $Q_2 = 12 - 2P_2 + P_1$ . They face same fixed cost of \$20 and variable cost is zero for both firms. The firms play a Bertrand game.

i) Derive and interpret the outcome of the game when firms produce differentiated goods

ii) Instead of choosing the prices independently what will happen if the two firms collude?

b) Explain the meaning of a Nash eq. when firms are competing with respect to price. Why is the equilibrium stable? Why don't the firms raise prices to the level that maximizes joint profit?

i) Given that,

Firm-1's demand  $Q_1 =$

Firm-2's demand  $Q_2$   
fixed cost

Now,

$$\pi_1 = P_1 Q_1 - 20$$

$$= P_1 (12 - 2P_1 + P_2) -$$

$$= 12P_1 - 2P_1^2 + P_1 P_2$$

FONC requires -

$$\frac{\partial \pi_1}{\partial P_1} = 12 - 4P_1 + P_2$$

$$\therefore \frac{\partial \pi_1}{\partial P_1} = 0$$

$$\Rightarrow 12 - 4P_1 + P_2 = 0$$

$$\Rightarrow P_2 = 4P_1 - 12$$

Again,

$$\pi_2 = P_2 Q_2 - 20 =$$

$$= 12P_2 - 2P_2^2 + P_1 P_2$$

$$\text{FONC requires, } \frac{\partial \pi_2}{\partial P_2}$$

i) Given that,

Firm-1's demand  $Q_1 = 12 - 2P_1 + P_2$

Firm-2's demand  $Q_2 = 12 - 2P_2 + P_1$  and

fixed cost = \$20,  $VC = 0$

Now,

$$\pi_1 = P_1 Q_1 - 20$$

$$= P_1 (12 - 2P_1 + P_2) - 20$$

$$= 12P_1 - 2P_1^2 + P_1 P_2 - 20$$

FONC requires —

$$\frac{\partial \pi_1}{\partial P_1} = 12 - 4P_1 + P_2$$

$$\therefore \frac{\partial \pi_1}{\partial P_1} = 0$$

$$\Rightarrow 12 - 4P_1 + P_2 = 0$$

$$\Rightarrow P_2 = 4P_1 - 12 \quad \text{--- (i)}$$

Again,

$$\pi_2 = P_2 Q_2 - 20 = P_2 (12 - 2P_2 + P_1) - 20 = 12P_2 - 2P_2^2 + P_1 P_2 - 20$$

$$= 12P_2 - 2P_1 P_2 + P_2^2 - 20$$

$$\text{FONC requires, } \frac{\partial \pi_2}{\partial P_2} = 0$$

$$12 - 4P_2 + P_1 = 0$$

$$\Rightarrow 12 - 4P_2 + P_1 = 0$$

$$\Rightarrow 12 - 4(4P_1 - 12) + P_1 = 0 \quad [\text{from (1)}]$$

$$\Rightarrow 12 - 16P_1 + 48 + P_1 = 0$$

$$\Rightarrow -15P_1 = -60$$

$$\Rightarrow P_1 = \frac{60}{15}$$

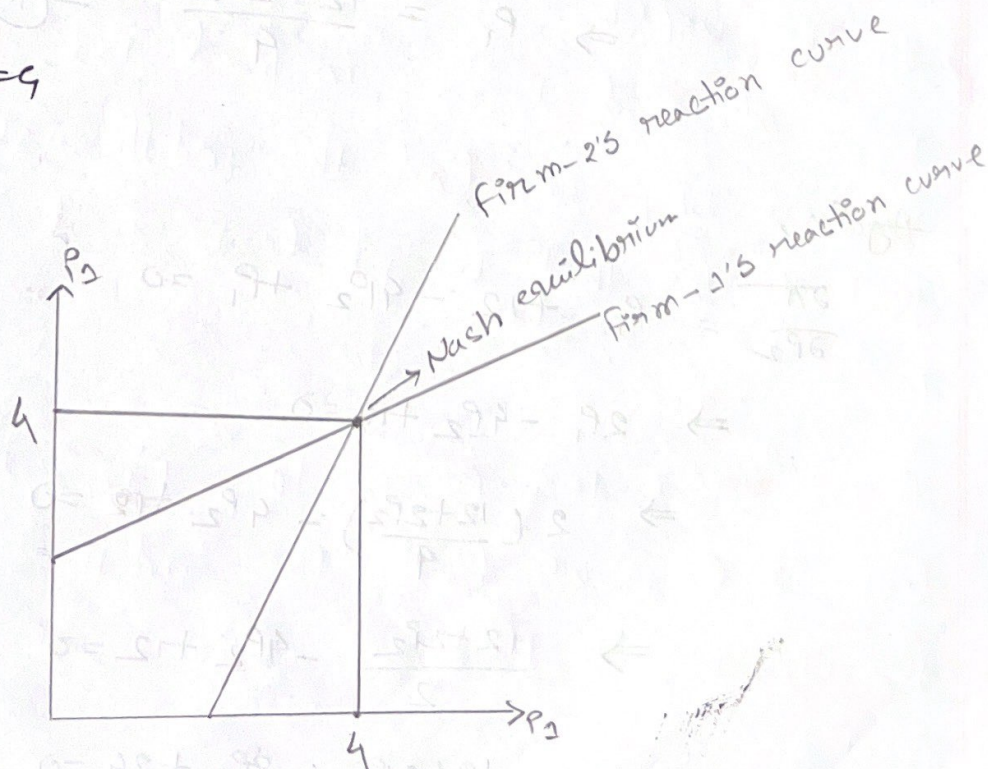
$$\therefore P_1 = 4$$

$$\therefore P_2 = 4P_1 - 12$$

$$= 4 \times 4 - 12$$

$$= 4$$

$$\therefore P_1 = P_2 = 4$$



Bartrand Model

ii) collusion solution

$$\pi = \pi_1 + \pi_2$$

$$= 12P_1 - 2P_1^2 + P_1P_2 - 20 + 12P_2 - 2P_2^2 + P_1P_2 - 20$$

$$= 12P_1$$

Now,

$$\frac{\partial \pi}{\partial P_1} = 12 - 4P_1 + P_2 + P_2 = 0$$

$$\Rightarrow 12 - 4P_1 + 2P_2 = 0$$

$$\Rightarrow 12 + 2P_2 = 4P_1$$

$$\Rightarrow P_1 = \frac{12 + 2P_2}{4} \quad \text{--- (i)}$$

Again,

$$\frac{\partial \pi}{\partial P_2} = P_1 + 12 - 4P_2 + P_1 = 0$$

$$\Rightarrow 2P_1 - 4P_2 + 12 = 0$$

$$\Rightarrow 2 \left( \frac{12 + 2P_2}{4} \right) - 4P_2 + 12 = 0$$

[from (i)]

$$\Rightarrow \frac{12 + 2P_2}{2} - 4P_2 + 12 = 0$$

$$\Rightarrow 12 + 2P_2 - 8P_2 + 24 = 0$$

$$\Rightarrow 12 - 6P_2 + 24 = 0$$

$$\Rightarrow 12 + 24 = 6P_2$$

$$\Rightarrow 36 = 6P_2$$

$$\therefore P_2 = \frac{12 + 2P_2}{4}$$

$$= \frac{12 + 2 \times 6}{4}$$

$$= 6$$

$$\therefore P_1 = P_2 = 6$$

$$\pi_1 = P_1 (12 - 2P_1) + P_1P_2$$

$$= 6 (12 - 2 \times 6) + 6 \times 6$$

$$= 36 - 20 = 16$$

$$\pi_2 = P_2 (12 - 2P_2) + P_1P_2$$



$$\Rightarrow 12 - 6P_2 + 24 = 0$$

$$\Rightarrow 12 + 24 = 6P_2$$

$$\Rightarrow 36 = 6P_2$$

$$\therefore P_2 = 6$$

$$\therefore P_1 = \frac{12 + 2P_2}{9}$$

$$= \frac{12 + 2 \times 6}{9}$$

$$= 6$$

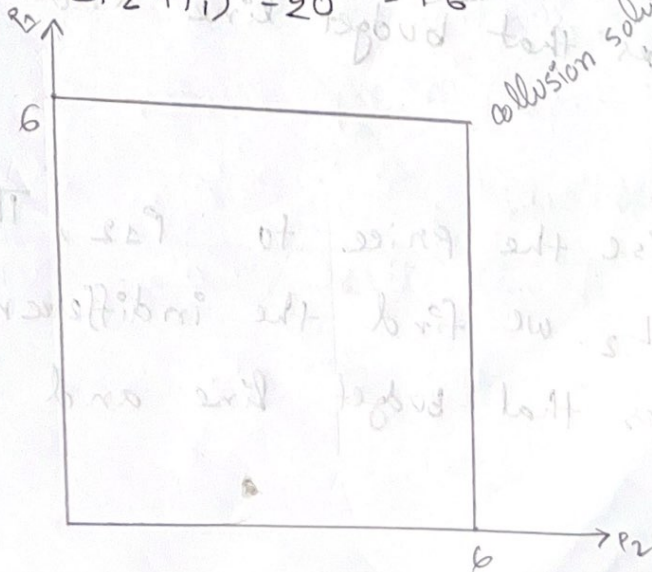
$$\therefore P_1 = P_2 = 6$$

$$\pi_1 = P_1(12 - 2P_1 + P_2) - 20$$

$$= 6(12 - 2 \times 6 + 6) - 20$$

$$= 36 - 20 = 16$$

$$\pi_2 = P_2(12 - 2P_2 + P_1) - 20 = 16$$



15. b) Illustrate and explain the followings for a normal good:

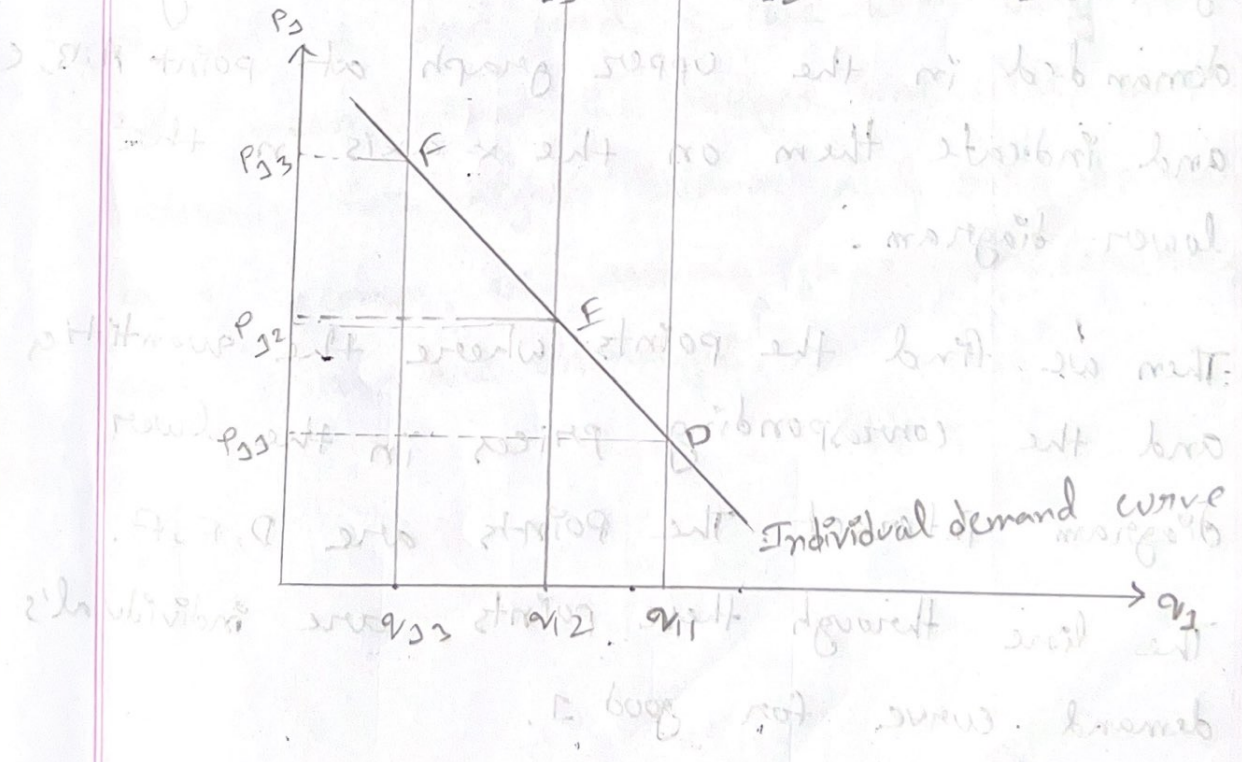
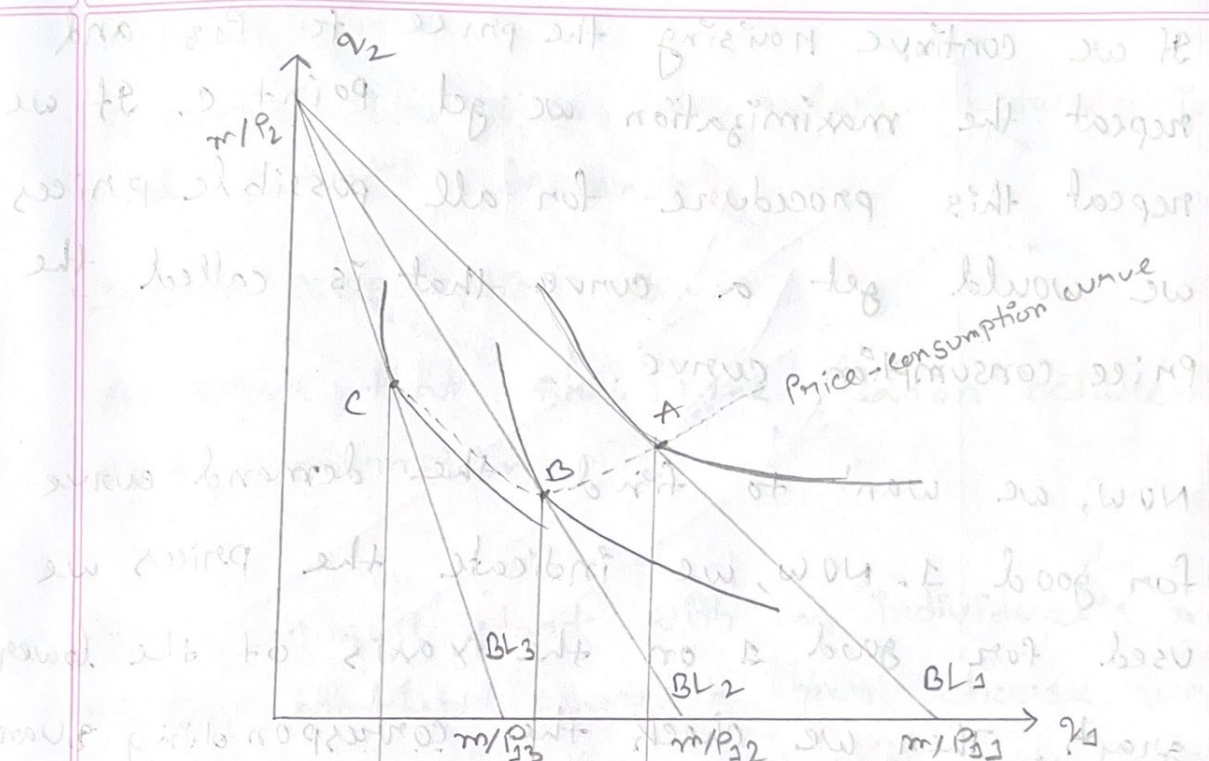
i) Derivation of individual demand curve from price consumption curve (PCC)

ii) Derivation of Engel curve from the Income consumption curve (ICC)

i) Price consumption curve is a curve that shows how a consumer chooses to consume at different prices.

Suppose we hold the price of good 2 constant. In the upper part of the figure, suppose the price of good 1 is initial  $P_{11}$ . Then the budget line is  $BL_1$ . We find the indifference curve that just touches that budget line and label the point A.

If we raise the price to  $P_{12}$ , Then the budget line is  $BL_2$ . We find the indifference curve that just touches that budget line and label the point B.



If we continue raising the price to  $P_3$  and repeat the maximization we get point c. If we repeat this procedure for all possible prices we would get a curve that is called the Price-consumption curve.

Now, we want to find the demand curve for good 1. Now, we indicate the prices we used for good 1 on the y-axis of the lower graph. Then we check the corresponding quantity demanded in the upper graph at point A, B, C and indicate them on the x-axis in the lower diagram.

Then we find the points where the quantities and the corresponding prices in the lower diagram intersect. The points are D, E, F.

The line through these points are individual's demand curve for good 1.

b) Engel curve:

A curve that shows and quantity demanded income consumption

A curve that shows and consumption.

Now, we will start maximization problem of good 1 and good 2. This means that the curve shifts outwards for higher incomes. As prices are un

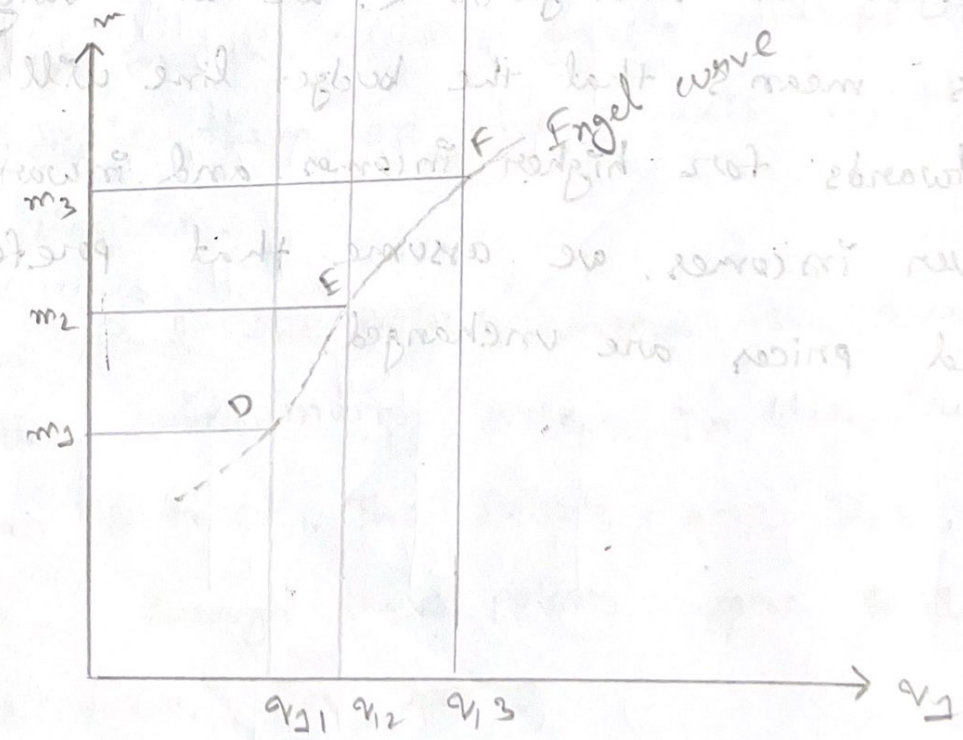
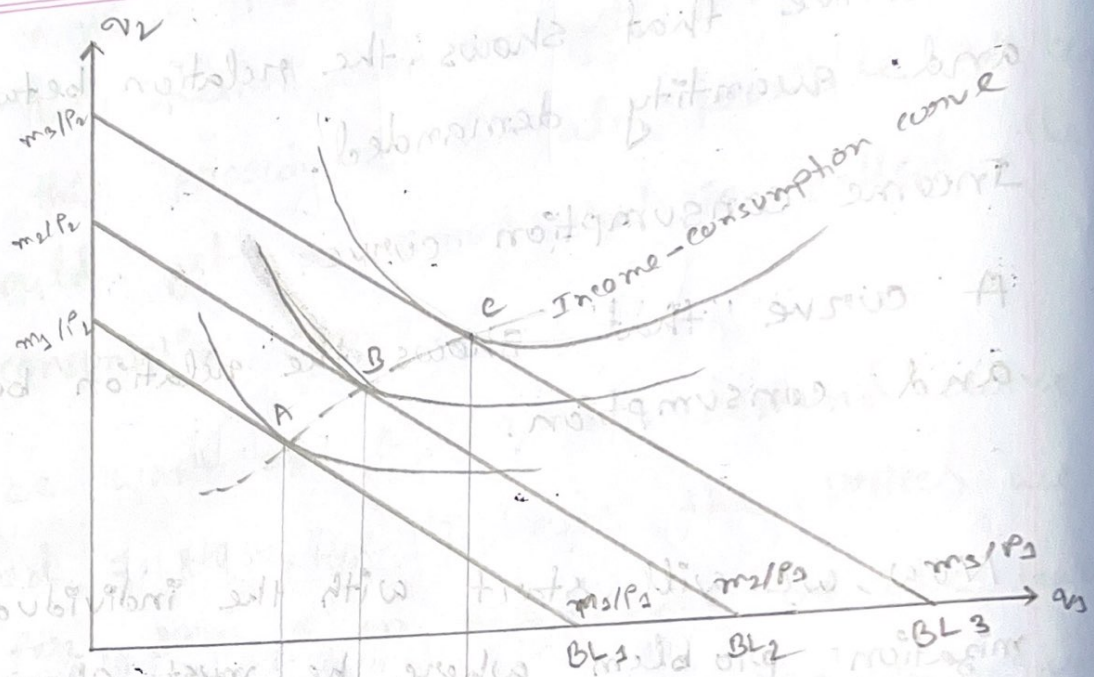
b) Engel curve:

A curve that shows the relation between income and quantity demanded.

Income consumption curve:

A curve that shows the relation between income and consumption.

Now, we will start with the individual's maximization problem where he must choose quantities of good 1 and good 2. We will vary income  $m$ . This means that the budget line will shift outwards for higher incomes and inwards for lower incomes. We assume that preferences ~~are~~ and prices are unchanged.



In the figure, for the increasingly higher incomes  $m_1$ ,  $m_2$  and  $m_3$ , the budget line becomes  $BL_1$ ,  $BL_2$  and  $BL_3$ . Then we find the utility maximization points for each budget line: A, B, C. If we do that for all possible incomes, we would get so-called income-consumption curve. This curve shows the optimal consumption of good 1 and good 2 at different incomes.

Now, we indicate the quantities corresponding to points A, B, C i.e.  $q_{11}$ ,  $q_{12}$ ,  $q_{13}$  in the below lower graph. Then we indicate the incomes  $m_1$ ,  $m_2$ ,  $m_3$  on Y-axis and we get a curve called Engel curve through the points D, E, F.

$$(i) \quad q_1 = \frac{VG}{PG}$$

$$(ii) \quad q_1 = \frac{VG}{PG}$$

$$(iii) \quad q_1 = \frac{VG}{PG}$$